

Mass dimension: 4

nD: 0, H: 2, H+: 2

SU2_W-Basis Tensors

$$(T_{SU2})_{ijkl}^0 = \varepsilon_{ik}\varepsilon_{jl} \quad (1)$$

$$(T_{SU2})_{ijkl}^1 = \varepsilon_{ij}\varepsilon_{kl} \quad (2)$$

H0H0H+0H+0

$$-\left(\frac{1}{2}\hat{\mu}_h^2\right)(T_{SU2})_{ijkl}^0 H_i H_j (H^\dagger)_k (H^\dagger)_l \quad (3)$$

$$+\left(\frac{1}{4}\hat{\mu}_h^2\right)(T_{SU2})_{ijkl}^1 H_i H_j (H^\dagger)_k (H^\dagger)_l \quad (4)$$

Mass dimension: 6

nD: 2, H: 2, H+: 2

SU2_W-Basis Tensors

$$(T_{SU2})_{ijkl}^0 = \varepsilon_{ik}\varepsilon_{jl} \quad (5)$$

$$(T_{SU2})_{ijkl}^1 = \varepsilon_{ij}\varepsilon_{kl} \quad (6)$$

H0H1H+0H+1

$$+\left(\frac{1}{2}\right)M_s^{-2}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}(T_{SU2})_{ijkl}^1 H_i D_{\alpha_1}^{\dot{\alpha}_1} H_j (H^\dagger)_k D_{\alpha_2}^{\dot{\alpha}_2} (H^\dagger)_l \quad (7)$$

H0H0H+1H+1

$$+\left(\frac{3}{4}\right)M_s^{-2}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}(T_{SU2})_{ijkl}^0 H_i H_j D_{\alpha_1}^{\dot{\alpha}_1} (H^\dagger)_k D_{\alpha_2}^{\dot{\alpha}_2} (H^\dagger)_l \quad (8)$$

$$-\left(\frac{1}{8}\right)M_s^{-2}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}(T_{SU2})_{ijkl}^1 H_i H_j D_{\alpha_1}^{\dot{\alpha}_1} (H^\dagger)_k D_{\alpha_2}^{\dot{\alpha}_2} (H^\dagger)_l \quad (9)$$

nD: 0, H: 3, H+: 3

SU2_W-Basis Tensors

$$(T_{SU2})_{ijklmn}^0 = \varepsilon_{il}\varepsilon_{jm}\varepsilon_{kn} \quad (10)$$

$$(T_{SU2})_{ijklmn}^1 = \varepsilon_{ik}\varepsilon_{jm}\varepsilon_{ln} \quad (11)$$

$$(T_{SU2})_{ijklmn}^2 = \varepsilon_{ik}\varepsilon_{jl}\varepsilon_{mn} \quad (12)$$

$$(T_{SU2})_{ijklmn}^3 = \varepsilon_{ij}\varepsilon_{km}\varepsilon_{ln} \quad (13)$$

$$(T_{SU2})_{ijklmn}^4 = \varepsilon_{ij}\varepsilon_{kl}\varepsilon_{mn} \quad (14)$$

H0H0H0H+0H+0H+0

$$-\left(\frac{1}{2}\lambda_h\right)M_s^{-2}(T_{SU2})^0_{ijklmn}H_iH_jH_k(H^\dagger)_l(H^\dagger)_m(H^\dagger)_n \quad (15)$$

$$+\left(\frac{1}{6}\lambda_h\right)M_s^{-2}(T_{SU2})^1_{ijklmn}H_iH_jH_k(H^\dagger)_l(H^\dagger)_m(H^\dagger)_n \quad (16)$$

$$+\left(\frac{1}{6}\lambda_h\right)M_s^{-2}(T_{SU2})^2_{ijklmn}H_iH_jH_k(H^\dagger)_l(H^\dagger)_m(H^\dagger)_n \quad (17)$$

$$+\left(\frac{1}{6}\lambda_h\right)M_s^{-2}(T_{SU2})^3_{ijklmn}H_iH_jH_k(H^\dagger)_l(H^\dagger)_m(H^\dagger)_n \quad (18)$$

$$-\left(\frac{1}{3}\lambda_h\right)M_s^{-2}(T_{SU2})^4_{ijklmn}H_iH_jH_k(H^\dagger)_l(H^\dagger)_m(H^\dagger)_n \quad (19)$$

nD: 0, H: 2, H+: 1, L+: 1, eC+: 1

SU2_W-Basis Tensors

$$(T_{SU2})^0_{ijkl} = \varepsilon_{ik}\varepsilon_{jl} \quad (20)$$

$$(T_{SU2})^1_{ijkl} = \varepsilon_{ij}\varepsilon_{kl} \quad (21)$$

H0H0H+0L+0eC+0

$$+\left(\frac{1}{2}\right)M_s^{-2}(y_e)^{uv}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}(T_{SU2})^0_{ijkl}H_iH_j(H^\dagger)_k(L^\dagger)^{\dot{\alpha}_1,u}_l(e^\dagger_{\mathbb{C}})^{\dot{\alpha}_2,v}_c \quad (22)$$

$$-\left(\frac{1}{4}\right)M_s^{-2}(y_e)^{uv}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}(T_{SU2})^1_{ijkl}H_iH_j(H^\dagger)_k(L^\dagger)^{\dot{\alpha}_1,u}_l(e^\dagger_{\mathbb{C}})^{\dot{\alpha}_2,v}_c \quad (23)$$

nD: 0, H: 2, H+: 1, Q+: 1, dC+: 1

SU3_C-Basis Tensors

$$(T_{SU3})^0_{abc} = +\left(\frac{1}{2}\right)\varepsilon_{abc} - \left(\frac{1}{2}\right)\varepsilon_{bac} \quad (24)$$

SU2_W-Basis Tensors

$$(T_{SU2})^0_{ijkl} = \varepsilon_{ik}\varepsilon_{jl} \quad (25)$$

$$(T_{SU2})^1_{ijkl} = \varepsilon_{ij}\varepsilon_{kl} \quad (26)$$

H0H0H+0Q+0dC+0

$$+\left(\frac{1}{4}\right)M_s^{-2}(y_d)^{uv}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}(T_{SU2})^0_{ijkl}(T_{SU3})^0_{abc}H_iH_j(H^\dagger)_k(Q^\dagger)^{\dot{\alpha}_1,u}_{l,ab}(d^\dagger_{\mathbb{C}})^{\dot{\alpha}_2,v}_c \quad (27)$$

$$-\left(\frac{1}{8}\right)M_s^{-2}(y_d)^{uv}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}(T_{SU2})^1_{ijkl}(T_{SU3})^0_{abc}H_iH_j(H^\dagger)_k(Q^\dagger)^{\dot{\alpha}_1,u}_{l,ab}(d^\dagger_{\mathbb{C}})^{\dot{\alpha}_2,v}_c \quad (28)$$

nD: 0, Q: 1, uC: 1, H: 2, H+: 1

SU3_ C-Basis Tensors

$$(T_{SU3})_{abc}^0 = +\left(\frac{1}{2}\right)\varepsilon_{abc} - \left(\frac{1}{2}\right)\varepsilon_{acb} \quad (29)$$

SU2_ W-Basis Tensors

$$(T_{SU2})_{ijkl}^0 = \varepsilon_{ik}\varepsilon_{jl} \quad (30)$$

$$(T_{SU2})_{ijkl}^1 = \varepsilon_{ij}\varepsilon_{kl} \quad (31)$$

Q0uC0H0H0H+0

$$-\left(\frac{1}{8}\right)M_s^{-2}(y_u^\dagger)^{vu}\varepsilon^{\alpha_1\alpha_2}(T_{SU2})_{ijkl}^0(T_{SU3})_{abc}^0Q_{\alpha_1,i,a}^u(u_{\mathbb{C}})_{\alpha_2,bc}^vH_jH_k(H^\dagger)_l \quad (32)$$

$$-\left(\frac{1}{8}\right)M_s^{-2}(y_u^\dagger)^{vu}\varepsilon^{\alpha_1\alpha_2}(T_{SU2})_{ijkl}^1(T_{SU3})_{abc}^0Q_{\alpha_1,i,a}^u(u_{\mathbb{C}})_{\alpha_2,bc}^vH_jH_k(H^\dagger)_l \quad (33)$$